

Homework 2

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Math 38

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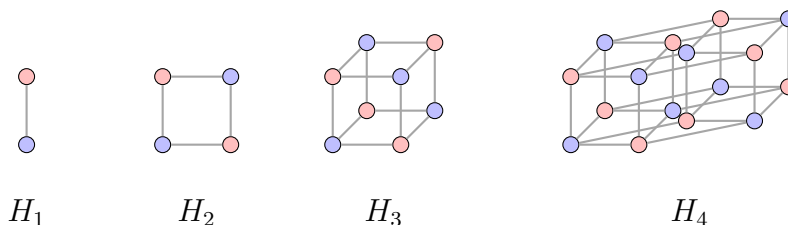
Question 1. Prove or disprove: All hypercubes H_n are bipartite.

Proof. We shall prove this statement by induction since a hypercube itself is constructed recursively. In a simple H_1 , it is trivial that the hypercube is bipartite, so we assume that the case holds for H_{k-1} . Now, consider a hypercube H_k , which can be constructed from a pair of bipartite hypercubes H^0 and H^1 , each isomorphic to H_{k-1} , by adding an edge between each vertex $v^0 \in V(H^0)$ and its corresponding vertex $v^1 \in V(H^1)$.

By our inductive hypothesis, there exists a valid 2-coloring for the vertices of H^0 using the colors blue and red. For the second copy H^1 , we define the color of each vertex v^1 based on its counterpart v^0 as follows:

$$\text{color}(v^1) = \begin{cases} \text{blue,} & \text{if color}(v^0) = \text{red} \\ \text{red,} & \text{if color}(v^0) = \text{blue} \end{cases}$$

This construction ensures that all edges within each H_{k-1} connect vertices of different colors. Specifically, if two vertices are adjacent in H^1 , their counterparts in H^0 were adjacent and had different colors; thus, their inverted colors in H^1 are also different. Furthermore, the edges added to connect H^0 to H^1 connect each vertex v^0 directly to v^1 . Since we assigned opposite colors by definition, H_k is 2-colorable and therefore bipartite. I have shown examples below:



□

Question 2. Let G be a connected simple graph not having any P_4 (path graph) or C_3 (cycle graph) as an induced subgraph. Prove that G is a complete bipartite graph.

Proof. First, since G has no induced C_3 , it is triangle-free. A graph is triangle-free if and only if it does not contain K_3 as a subgraph. Since G is a simple graph with no C_3 , it must be bipartite. If it were not bipartite, it would contain an odd cycle. If the shortest odd cycle is C_k with $k \geq 5$, any two non-adjacent vertices on that cycle would form an induced P_4 or a chord (creating a smaller cycle), contradicting the induced P_4 and C_3 constraints. Thus, G is bipartite.

Let the bipartition of $V(G)$ be (A, B) , where A and B are independent sets. Since G is connected, both A and B are non-empty (assuming $|V| \geq 2$). To show G is a complete bipartite graph, we must prove that every vertex $a \in A$ is adjacent to every vertex $b \in B$. Suppose for the sake of contradiction that there exist $a \in A$ and $b \in B$ such that $ab \notin E(G)$. Because G is connected, there exists a path between a and b . Let $P = (a, v_1, v_2, \dots, v_k, b)$ be the shortest path between them. Since G is bipartite and a, b are in different partitions, the path must have an odd number of edges (and an even number of vertices).

- If the shortest path has length 3, let the path be (a, b', a', b) . Since a and a' are in A , they are not adjacent. Similarly, b' and b are in B and not adjacent. If ab is not an edge, the path (a, b', a', b) is an induced path. This is a P_4 , which contradicts the assumption that G has no induced P_4 .
- If the shortest path has length $k \geq 5$, any four consecutive vertices in the path would form an induced P_4 . If they were not induced (i.e., there was a chord), it would contradict the "shortest path" assumption or create a C_3 .

Therefore, no such non-adjacent pair (a, b) can exist. Every vertex in A must be adjacent to every vertex in B . Thus, G is a complete bipartite graph $K_{m,n}$. $\square$

Question 3.

- (a) Show that an even graph has no cut-edge.

Proof. Suppose for the sake of contradiction that G is an even graph that contains a cut-edge $e = uv$. By the definition of a cut-edge, the removal of e increases the number of connected components of G . Let G_1 and G_2 be the two disjoint components created by the removal of e , such that $u \in V(G_1)$ and $v \in V(G_2)$.

Now, consider the sum of the degrees of the vertices in the component G_1 .

- For any vertex $w \in V(G_1)$ such that $w \neq u$, its degree in G_1 is identical to its degree in G . Since G is an even graph, $d_{G_1}(w)$ is even.
- For the vertex u , its degree in G_1 is $d_G(u) - 1$. Since $d_G(u)$ is even, $d_{G_1}(u)$ must be odd.

By the Handshaking Lemma, the sum of the degrees of all vertices in any graph must be even:

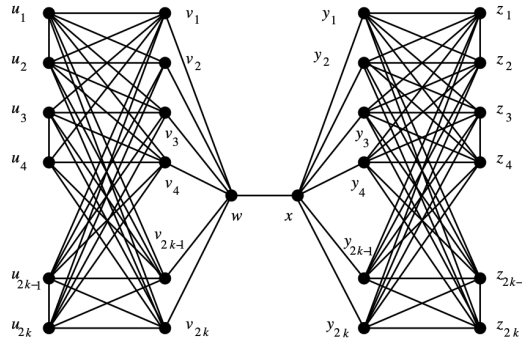
$$\sum_{x \in V(G_1)} d_{G_1}(x) = 2|E(G_1)|$$

However, the sum of the degrees in G_1 consists of one odd value ($d_{G_1}(u)$) and a set of even values. The sum of a single odd number and any number of even numbers is always odd. This implies that the sum of degrees in G_1 is odd, which directly contradicts the Handshaking Lemma. Therefore, our initial assumption that G contains a cut-edge must be false. Thus, an even graph has no cut-edge. $\square$

- (b) For any $k \geq 1$, construct a $(2k + 1)$ -regular simple graph with a cut-edge.

Solution. Source: IIT

(<https://cse.iitkgp.ac.in/~abhij/course/theory/GTH/Spring03/soln1.pdf>) $\square$



Question 4. A sequence of $(d_1, \dots, d_n)$ non-increasing (i.e. $d_1 \geq d_2 \geq \dots \geq d_n$) is a degree sequence of a loopless graph iff its sum is even and $d_1 \leq d_2 + \dots + d_n$.

Proof. ($\implies$) Suppose $(d_1, \dots, d_n)$ is the degree sequence of a loopless graph $G = (V, E)$. First, we observe that the sum of the degrees must be even. By the Handshaking Lemma, we know that

$$\sum_{i=1}^n d_i = 2|E|.$$

Since the sum is equal to twice the number of edges, it is therefore an even integer. Now, we must show that $d_1 \leq d_2 + \dots + d_n$. In a loopless graph, every edge incident to the vertex of maximum degree, v_1 , must connect to some other vertex v_j where $j \neq 1$. Each such edge contributes exactly 1 to the value of d_1 and contributes at least 1 to the sum of the remaining degrees $\sum_{i=2}^n d_i$.

More formally, let $E(v_1, V - \{v_1\})$ be the set of edges incident to v_1 . Since there are no loops, all d_1 edges incident to v_1 must have their other endpoint in the set $\{v_2, \dots, v_n\}$. Thus:

$$d_1 = |E(v_1, V - \{v_1\})|$$

The total sum of the degrees of the remaining vertices is

$$\sum_{i=2}^n d_i = |E(v_1, V - \{v_1\})| + 2|E(V - \{v_1\})|,$$

where $E(V - \{v_1\})$ represents the edges that do not involve v_1 . This simplifies to

$$\sum_{i=2}^n d_i = d_1 + 2|E(V - \{v_1\})|.$$

Since the number of edges $2|E(V - \{v_1\})| \geq 0$, it follows immediately that $d_1 \leq \sum_{i=2}^n d_i$.

($\impliedby$) Assume the sum $\sum d_i$ is even and $d_1 \leq \sum_{i=2}^n d_i$. We proceed by induction on the sum $S = \sum d_i$.

The base case $S = 0$ is trivial; a graph with n isolated vertices realizes the sequence. Since the sum must be even, the next case is $S = 2$. The condition $d_1 \leq \sum_{i=2}^n d_i$ implies $d_1 = 1$, so the sequence must be $(1, 1, 0, \dots, 0)$, which is realized by a single edge between v_1 and v_2 . Now assume any valid sequence with sum $k < S$ is realizable. Consider a sequence with sum S . We split this into two cases:

Case 1: $d_1 = d_2 + \dots + d_n$. We can realize this by drawing d_j edges between v_1 and v_j for each $j \in \{2, \dots, n\}$. Since d_1 is exactly the sum of the others, v_1 will have exactly d_1 edges incident to it, and each v_j will have exactly d_j .

Case 2: $d_1 < d_2 + \dots + d_n$. Let d_i and d_{i+1} be the last two non-zero entries in the sequence (where $i + 1 \geq 2$). We form a new sequence d' by subtracting 1 from both:

$$d' = (d_1, d_2, \dots, d_i - 1, d_{i+1} - 1, 0, \dots, 0)$$

We check the two inductive conditions for d' :

1. **Even sum:** Since we subtracted exactly 2 from the total sum, the new sum $S - 2$ is still even.
2. **Inequality:** We need $d'_1 \leq \sum_{j=2}^n d'_j$. Since $d'_1 = d_1$ and we only reduced degrees in the summation part, we must ensure $d_1 \leq (d_2 + \cdots + d_n) - 2$. However, because S and d_1 have the same parity (since $S - d_1 = \sum_{j=2}^n d_j \geq d_1$), the difference $(d_2 + \cdots + d_n) - d_1$ must be an even integer ≥ 0 . Since we are in Case 2 ($d_1 < \sum_{j=2}^n d_j$), the difference is at least 2. Thus, $d_1 \leq (\sum_{j=2}^n d_j) - 2$ holds.

By the inductive hypothesis, d' is realized by a graph G' . We obtain G by adding an edge between v_i and v_{i+1} in G' . This increases their degrees by 1, realizing the original sequence d . □

Question 5. Prove or disprove: If G is an n -vertex simple graph with maximum degree $\lceil n/2 \rceil$ and minimum degree $\lfloor n/2 \rfloor - 1$, then G is connected.

Proof. Suppose for contradiction that G is disconnected. Let C_1 and C_2 be two connected components of G , with $|C_1| = p$ and $|C_2| = q$, where $p + q \leq n$ and $p \leq q$ without loss of generality.

Since C_1 and C_2 are components, no vertex in C_1 is adjacent to any vertex in C_2 . Thus any vertex $v \in C_1$ can have degree at most $p - 1$, i.e. $d(v) \leq p - 1$. Now, the vertex of maximum degree lies in the larger component C_2 , so $\Delta(G) \leq q - 1 \leq n - p - 1$. But by hypothesis, $\Delta(G) = \lceil n/2 \rceil$, so

$$\lceil n/2 \rceil \leq n - p - 1 \implies p \leq n - 1 - \lceil n/2 \rceil = \lfloor n/2 \rfloor - 1.$$

Thus every vertex in C_1 satisfies $d(v) \leq p - 1 \leq \lfloor n/2 \rfloor - 2$. But this contradicts the hypothesis that the minimum degree of G is $\lfloor n/2 \rfloor - 1$. Therefore G must be connected. $\square$